

Backtesting Probability of Default Models

The Multi-Period Average Testing

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Internal Ratings-Based Modeling Process

- The development of an Internal Ratings-Based (IRB) model consists of two main areas: the development of the risk differentiation function and risk quantification.
- Development of the risk differentiation function is the first step in the overall model development process. The main goal of this step is to ensure the model can adequately rank risk, providing meaningful differentiation between low- and high-risk obligors or facilities.
- After developing the risk differentiation function, practitioners perform the risk quantification process, which essentially involves assigning an appropriate risk level to the output of the risk differentiation function.
- Risk quantification typically consists of specific steps such as: assignment of the base or best estimate, quantification of appropriate adjustments (AA), quantification of the margin of conservatism (MoC), and quantification of any other additional conservatism, for instance that required by certain local regulators. The first two steps are typically referred to as calibration.
- These steps apply to all three risk parameters: Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD).

Probability of Default Backtesting

- As risk quantification is a regulatory requirement, practitioners pay significant attention to complying with these requirements.
- In addition to adhering to the prescribed risk quantification process, practitioners are required to assess the appropriateness of model estimates. This is typically referred to as backtesting.
- The primary purpose of backtesting in this context is to ensure that model estimates accurately predict the occurrence of defaults, i.e., that PD estimates provide reliable forecasts of default rates.
- In practice, this is one of the most comprehensive validation exercises, typically investigating model estimates across different portfolio subsegments and time dimensions.
- One commonly applied approach is to backtest estimates over a recent period of three to five years.
- In practice, for the purpose of testing PD on recent samples (three to five years), practitioners often employ a z-score test.
- This test relies on a continuous approximation of the binomial proportion distribution.

Probability of Default Backtesting cont.

- The z-score test is traditionally used in both one-sided and two-sided versions, often to assess evidence of underestimation or the accuracy (sometimes referred to as precision) of the PD estimate compared to the observed default rate.
- In practice, this multi-period average testing is typically performed by preparing the inputs - the observed default rate and the number of exposures - either by:
 - 1 calculating the observed default rate as a default-weighted average, with the total number of exposures equal to the sum of exposures across individual years;
 - 2 calculating the observed default rate as an arithmetic average of yearly default rates, with the number of exposures calculated in the same way.
- Given that the PD best estimate is typically defined as the arithmetic average of yearly default rates, the question arises as to whether the methods prevalent in practice properly prepare inputs for testing purposes. In other words, whether these testing approaches are statistically sound and whether they properly reflect the properties of the estimator.
- The purpose of this presentation is to demonstrate the calculation of the average default rate and the effective sample size as the main inputs for multi-period average PD testing using the z-score test. In simple terms, the proposed method aligns the calculation of inputs with the properties of the average default rate estimator and with the most commonly used method for calibrating PD best estimates.

The Z-Score Test

The test statistic of the z-score test is given as follows:

$$z_{score} = \frac{ODR - PD}{\sqrt{\frac{PD(1-PD)}{N}}}$$

where:

- ODR is the observed default rate;
- PD is the calibrated PD;
- N is the number of exposures.

Under the assumption that the z_{score} test statistic follows a standard normal distribution, a p-value is calculated accordingly.

Multi-Period Estimator

- When applying the z-score test, practitioners often assess the accuracy of the PD best estimate (two-sided test) or the underestimation of the final PD estimate (one-sided test), where the final estimate includes all add-ons on top of the best estimate.
- Typically, the PD best estimate is calculated as the arithmetic average of yearly default rates over the calibration period. Formally, $PD = \frac{1}{T} \sum_{i=1}^T ODR_i$, where T is the number of reference dates and ODR_i is the observed default rate at reference date i .
- Given this definition of the PD best estimate, practitioners should align the z-score test inputs for multi-year average testing with the estimator of the multi-year observed default rate. In simple terms, the multi-year average default rate should align with the PD best estimate approach, while the sample size N should align with the variance of the estimator.

Multi-Period Estimator cont.

- The average default rate used for z-score testing is straightforward to calculate, while the effective sample size needs to be aligned with the variance of the estimator.
- By equating the variance of the true average default rate, given by $\frac{\overline{ODR}(1-\overline{ODR})}{N_e}$, with the variance of the estimator, given by $\text{Var}(\frac{1}{T} \sum_{i=1}^T ODR_i)$, and under the assumption of independence of default rates across reference dates, it can be shown that the effective sample size is $N_e = \frac{T^2}{\sum_{i=1}^T \frac{1}{n_i}}$, where T is the number of reference dates used to calculate the average ODR and n_i is the number of exposures at each reference date.
- Practitioners should note that the above formula for the effective sample size N_e can be further extended to incorporate the dependency structure of ODR over time, which can be a significant factor even for yearly ODRs, and even more so for ODRs observed at higher frequencies, such as quarterly or monthly.

Simulation Study

The purpose of this simulation is to compare the results of multi-period PD testing between commonly used methods in practice and the approach proposed in this presentation, which is based on the effective sample size.

Like in a standard backtesting exercise, the main goal is to compare the calibrated PD with the multi-period average ODR using the z-score test.

For the purpose of demonstrating the differences, this simulation considers the following simplified scenario:

A PD best estimate (p_d) of 2% is backtested against the ODR using the z-score test. The ODR consists of defaults over two years (T). In the first year, the number of observed defaults is 15, and in the second year, it is 60 (d), out of 1,000 and 2,000 exposures (n) per year, respectively.

Finally, to demonstrate differences in the test outcome, the average ODR and the number of exposures used for this purpose are calculated using different methods.

R Example

```
#z-score test
zscore.test <- function(pdc, odr, no, alternative) {
  se <- sqrt((pdc * (1 - pdc)) / no)
  test.stat <- (odr - pdc) / se
  p.val <- switch(alternative,
    "two.sided" = 2 * min(pnorm(test.stat), 1 - pnorm(test.stat)),
    "less" = pnorm(test.stat, lower.tail = TRUE),
    "greater" = pnorm(test.stat, lower.tail = FALSE))
  return(p.val)
}

#simulation inputs
pd <- 0.02           #pd best estimate
T <- 2               #number of years (reference dates)
d <- c(15, 60)       #number of defaults per year
n <- c(1000, 2000)   #number of exposures per year
```

R Example cont.

#z-score test: default-weighted ODR and total number of exposures as the sum of individual years

```
zscore.test(pdc = pd,  
            odr = sum(d) / sum(n),  
            no  = sum(n),  
            alternative = "two.sided")
```

```
## [1] 0.05044725
```

#z-score test: yearly average ODR and average number of exposures

```
zscore.test(pdc = pd,  
            odr = mean(d / n),  
            no  = mean(n),  
            alternative = "two.sided")
```

```
## [1] 0.4891859
```

#z-score test: yearly average ODR and effective sample size

```
N.e <- T^2 / sum(1 / n)  
zscore.test(pdc = pd,  
            odr = mean(d / n),  
            no  = N.e,  
            alternative = "two.sided")
```

```
## [1] 0.3564561
```

Python Example

```
import numpy as np
from scipy.stats import norm

#z-score test
def zscore_test(pdc, odr, no, alternative):
    se = np.sqrt((pdc * (1 - pdc)) / no)
    test_stat = (odr - pdc) / se
    if alternative == "two_sided":
        p_val = 2 * min(norm.cdf(test_stat), 1 - norm.cdf(test_stat))
    elif alternative == "less":
        p_val = norm.cdf(test_stat)
    elif alternative == "greater":
        p_val = 1 - norm.cdf(test_stat)
    else:
        raise ValueError("Invalid alternative hypothesis")

    return p_val

#simulation inputs
pd = 0.02                #pd best estimate
T = 2                    #number of years (reference dates)
d = np.array([15, 60])  #number of defaults per year
n = np.array([1000, 2000]) #number of exposures per year
```

Python Example cont.

#z-score test: default-weighted ODR and total number of exposures as the sum of individual years

```
zscore_test(pdc = pd,  
            odr = np.sum(d) / np.sum(n),  
            no = np.sum(n),  
            alternative = "two_sided")
```

```
## np.float64(0.05044725207774947)
```

#z-score test: yearly average ODR and average number of exposures

```
zscore_test(pdc = pd,  
            odr = np.mean(d / n),  
            no = np.mean(n),  
            alternative = "two_sided")
```

```
## np.float64(0.48918594287319106)
```

#z-score test: yearly average ODR and effective sample size

```
N_e = T**2 / np.sum(1 / n)  
zscore_test(pdc = pd,  
            odr = np.mean(d / n),  
            no = N_e,  
            alternative = "two_sided")
```

```
## np.float64(0.3564561327439)
```